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Design and Simulation of Two-Wheel Balancing Robots

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***Abstract*-This paper addresses the limitations of current unmanned delivery vehicles in complex environments and harsh conditions such as disaster relief operations. By integrating the advantages of wheel-legged robots, this study proposes a modular, lightweight, and reconfigurable design to enhance adaptability in heterogeneous terrains. The modular approach facilitates maintenance and interchangeability in complex scenarios, while the lightweight, reconfigurable design improves traversability. Multi-vehicle collaborative operations expand payload capacity and functional versatility, offering a more efficient solution for logistics. For motion control, this research implements LQR algorithms and TMSLIP models to achieve basic motion control. D'Alembert's principle is utilized for joint action mapping, complemented by a velocity and acceleration-based strategy for jump phase switching. These approaches provide theoretical support for the wheel-legged robot's fundamental movements and jumping actions. Experimental results demonstrate the prototype's excellent performance in balance stability and jumping motions, validating the effectiveness of the proposed motion control strategies and jump planning methods. This study confirms the reliability and efficacy of wheel-legged robots in practical applications, particularly in challenging environments.**

Keywords - Intelligent delivery; Bipedal robot; modularization; TMSLIP model

I. Introduction

The rapid advancement of artificial intelligence and robotics over the past decade has led to the widespread integration of robots across various sectors, from daily life to industrial manufacturing and military applications. Within the field of robotics, wheeled and bipedal robots have garnered significant attention due to their unique advantages. However, as societal demands evolve, single-structure robot designs are increasingly inadequate. Consequently, hybrid robots incorporating multiple locomotion structures have emerged as a prominent research focus [1-4]. Among these, wheel-legged robots represent an innovative design that combines the advantages of both wheeled and legged structures [5-6]. These robots can adaptively switch locomotion modes based on environmental requirements, demonstrating excellent performance in both structured and unstructured terrains. This versatility significantly expands the potential applications of robots and promises to substantially enhance human work efficiency and productivity [7-9].

Against the backdrop of rapid developments in intelligent manufacturing and robotics, research on wheel-legged robots has made significant strides. Their superior traversability, adjustable ground clearance, and modular design confer unique advantages in fields such as logistics and delivery. Compared to traditional drone delivery systems, wheel-legged robots can carry larger and heavier payloads, effectively addressing load capacity limitations. Moreover, their modular design facilitates product upgrades and rapid repairs, particularly suitable for environments with limited maintenance conditions, such as mountainous regions. The compact design and standardized packaging also greatly improve efficiency in storage, transportation, and deployment. Notably, wheel-legged robots not only show promise in logistics but also demonstrate immense potential in specialized environments such as rescue operations and exploration. Their ability to adapt to complex terrains enables them to perform hazardous tasks during natural disasters like earthquakes and floods, significantly reducing the risk of human casualties [10].

In conclusion, wheel-legged robots, as a product of innovative fusion, are set to drive technological innovation in the logistics industry, improve delivery efficiency, and play crucial roles across broader fields, contributing significantly to enhancing overall social operational efficiency and quality of life. This study aims to delve into the design principles, control strategies, and applications of wheel-legged robots in complex environments, providing new insights and practical experience for the advancement of this cutting-edge field. By exploring the potential of wheel-legged robots, this research contributes to the ongoing evolution of robotics and its applications in addressing complex societal challenges.

II. Technology and Product

A. Research on Structure Design

The intelligent logistics delivery robot employs a wheel-leg design, offering enhanced mobility and stability to navigate complex terrains and obstacles, thereby improving the efficiency and automation of logistics delivery systems. A key feature is its modular design, comprising essential components such as a cargo compartment, support frame, electrical enclosure, drive module, and auxiliary wheels, allowing for interchangeability between different power units. To achieve advanced functionality while maintaining structural integrity and mechanical performance, the robot design demands a sophisticated approach to weight reduction. This project focuses on researching and developing each subassembly, utilizing finite element analysis tools and physical prototype testing. The research encompasses critical aspects including material selection, innovative quick-release connectors, integrated wheel-drive units, structural design optimization, novel manufacturing processes, and rigorous performance validation. Additionally, the project studies failure modes of lightweight materials under harsh environmental conditions. The ultimate goal is to synthesize these findings into an optimal configuration that meets logistics transportation demands, ensuring the development of a robust, efficient, and adaptable robot capable of operating in diverse and challenging environments.

The intelligent logistics delivery robot adopts an innovative wheel-leg design, combining a single-degree-of-freedom multi-link structure with a coupled system of high-performance joint motors and hub motors. This design allows for intelligent adjustment of ground clearance, enabling the robot to adapt flexibly to heterogeneous terrains. Based on varying ground clearance, the robot possesses two fundamental modes: crawling and standing, as illustrated in Figure 1.

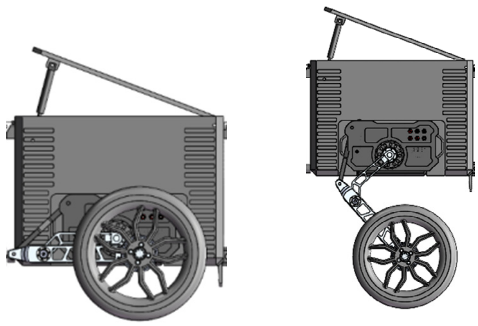

Figure 1. Intelligent logistics delivery robot wheel leg diagram.

B. Research on Motion Control

This research addresses the complex motion control processes for intelligent logistics delivery robots, focusing on fundamental movements such as balancing, forward motion, and turning. Our approach involves developing effective control strategies based on the robot's overall kinematic model. Specifically, we calculate reference feedback gain values as inputs for the Linear Quadratic Regulator (LQR) control algorithm. Sensor data is used as model state variables, providing real-time information about the robot's position, orientation, and dynamics. To manage non-linear dynamics, we linearize the system's equations through Taylor expansion, retaining the first-order terms to form the state-space equation.

In the combined vehicle body, we simplify the robot into a Wheeled Inverted Pendulum (WIP) linear model, assuming no slippage between the drive wheels and the ground, and rigid connections between body compartments. The front and rear body compartments are assumed to move vertically relative to the central robot section. Based on this, sensor data from the central section serves as the control input, allowing us to simplify the combined vehicle body into a four-degree-of-freedom dynamic model, as illustrated in Figure 2.

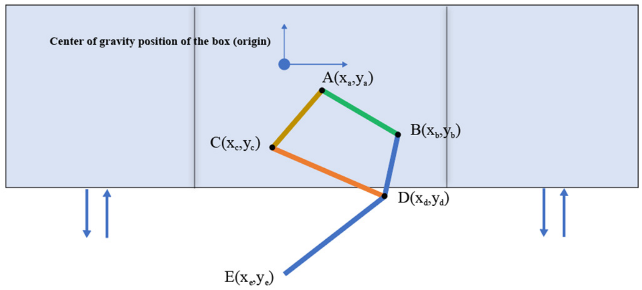

Figure 2. Schematic diagram of the four degree of freedom dynamic model of the combined vehicle body

C. Research on Optimization Design of Power Unit

This research investigates the optimization of the power unit for intelligent logistics delivery robots, with a focus on collaborative control methods for the wheel-assisted drive unit. The robot is modeled as a two-wheel distributed drive balance vehicle, where motors drive the wheels through reduction mechanisms and transmission devices. Steering is facilitated by electronic differential and torque coordination control. Precise control is achieved by analyzing motor control algorithms and switching motor operating modes, utilizing Hall sensors for feedback. The system's robustness and timeliness are enhanced through Field-Oriented Control (FOC) of the motors and CAN bus communication. Key motor design features include an axial flux motor structure to support high current loads, high peak power for achieving high rotational speeds and torque, new stator and rotor materials coupled with an innovative cooling structure for high power density and efficiency, and waterproof seals and coatings achieving an IP65 rating.

Parallel research is conducted on motor control algorithm optimization, battery management system design, and real-time monitoring and adjustment of vehicle dynamic parameters, focusing on system performance optimization and energy management. Machine learning concepts are employed to construct a motor control strategy training model, incorporating real-time data processing to mitigate the impact of initial system conditions and enhance the applicability and robustness of State of Charge (SOC) estimation. By combining model-based methods with data-driven approaches, motor control algorithm research utilizes reference trajectories generated by the model to inform algorithm optimization. Additionally, an equivalent model of dynamic characteristics is established to determine the optimal economic operating curve for the motor, laying the foundation for system configuration stability and motion control stability research. This comprehensive approach ensures that the power unit of the intelligent logistics delivery robot is optimized for high performance, efficiency, and reliable operation in various logistics scenarios.

D. Research on Trajectory Planning and Control

This research investigates the optimization of the power unit for intelligent logistics delivery robots, with a focus on collaborative control methods for the wheel-assisted drive unit. The robot is modeled as a two-wheel distributed drive balance vehicle, where motors drive the wheels through reduction mechanisms and transmission devices. Steering is facilitated by electronic differential and torque coordination control. Precise control is achieved by analyzing motor control algorithms and switching motor operating modes, utilizing Hall sensors for feedback. The system's robustness and timeliness are enhanced through Field-Oriented Control (FOC) of the motors and CAN bus communication. Key motor design features include an axial flux motor structure to support high current loads, high peak power for achieving high rotational speeds and torque, new stator and rotor materials coupled with an innovative cooling structure for high power density and efficiency, and waterproof seals and coatings achieving an IP65 rating.

This research on trajectory planning and control draws inspiration from the skeletal structures and movement patterns of animals in nature. We have refined the jumping process into four distinct phases: take-off, ascending flight, descending flight, and landing, analyzed using the simplified Two-Mass Spring-Loaded Inverted Pendulum (TMSLIP) model. The control strategy innovatively decouples jumping, velocity, and posture. The robot is simplified into a virtual model, with movements analyzed and planned, then mapped onto the robot's joint space. The legs, modeled as mass-less virtual springs attached beneath a mass point, provide jumping capability as per the Spring-Loaded Inverted Pendulum (SLIP) model. This simplification reduces computational load and avoids complex dynamic calculations. Using Recurdyn simulation software, a dynamic simulation model is constructed, aligning leg dimensions and joint positions with the engineering design. The model's weight matches a fully loaded 85kg robot with a limb module carrying a 60kg payload. Springs between the "thighs" and the chassis reduce motor load during standing balance and offer suspension for vibration reduction. The constructed dynamic simulation model is illustrated in Figure 3.

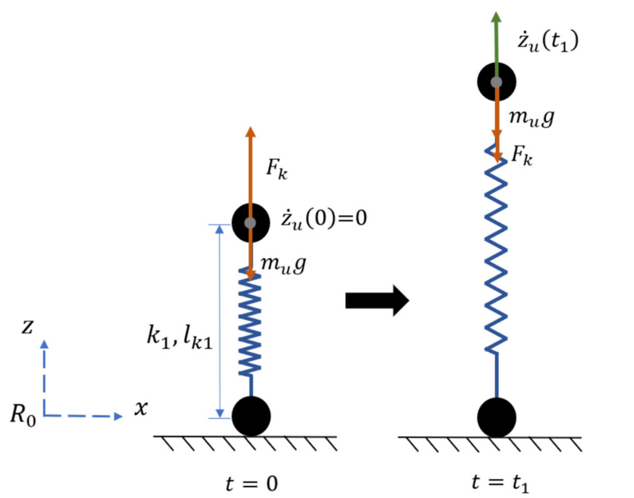

Figure 3. Schematic diagram of jumping dynamics model of intelligent logistics distribution robot.

E. System integration

The control system for this project is implemented using Ubuntu 20.04, with EtherCAT as the communication protocol, providing us with extensive experience in developing electric drive, electronic control, and sensor systems for robotics. Our research spans several key areas: we utilized combined VMC (Virtual Model Control) and QP (Quadratic Programming) algorithms to investigate gait and steering control of legged robots, enhancing their locomotion efficiency and maneuverability. We also focused on a dual-wheel self-balancing vehicle system, exploring drive control, differential steering, and stability control through distributed electric drive systems to optimize dynamics and maintain balance across various conditions. Additionally, we conducted compliance analysis using MPC (Model Predictive Control) for force feedback during a robot's stance phase, aiming to improve safety and compliance. Path planning, obstacle detection, and avoidance were advanced using Lidar and GPS technologies to ensure efficient and safe navigation. These research methods and outcomes provide valuable insights into variable structure precision control, instability, self-recovery mechanisms, and the dual-mode locomotion stability of wheeled and legged robots, contributing significantly to advancements in robotic control systems.

III. PRODUCT SUPERIORITYCODE MODULES

A. Motor Control and Power Unit

This section simulates motor control of a two-wheel self-balancing vehicle, achieving precise steering through electronic differential and torque coordination control.

```
% Motor Control and Power Unit
clear all; clc;
% Set robot parameters
wheel_radius = 0.15; % Wheel radius (m)
wheel_base = 0.5; % Distance between wheels (m)
motor_max_torque = 5; % Maximum motor torque (Nm)
motor_max_speed = 100; % Maximum motor speed (rad/s)
% Set target speed and steering angle
target_velocity = 1.0; % Target linear speed (m/s)
target_angle = 15; % Target steering angle (degrees)
% Calculate wheel speeds
target_angle_rad = deg2rad(target_angle);
v_left = (2 * target_velocity - target_angle_rad * wheel_base) / (2 * wheel_radius);
v_right = (2 * target_velocity + target_angle_rad * wheel_base) / (2 * wheel_radius);
% Limit speeds to within maximum motor speed range
v_left = min(motor_max_speed, max(-motor_max_speed, v_left));
v_right = min(motor_max_speed, max(-motor_max_speed, v_right));
disp(['Left wheel speed: ', num2str(v_left), ' rad/s']);
disp(['Right wheel speed: ', num2str(v_right), ' rad/s']);
```

B. Program flowchart

This project focuses on the study of a modular, composable dual-wheel legged robot, starting with the development of multi-order kinematic equations from a single robot to combinations involving multiple robots. By analyzing the system characteristics, we aim to establish a system model under variable dynamic constraints. We are innovating algorithms based on the inverted pendulum model at the foot end and the kinematic model of the robot's legs to achieve flexible control of the prototype. Our research also includes the estimation of state of charge (SOC) and the economic drive of motors, which are crucial for sustainable and efficient operation. Moreover, we pursue innovative structural designs for the robot, guided by specific design inputs. In addition, we build filtering models based on the impedance characteristics of sensors to study the stability and motion control of the dual-wheel legged robot. The detailed research content, along with the logical sequence and interrelations of each part of the study, is illustrated in Figure 4. This comprehensive approach ensures that each aspect of design and control is interwoven, maximizing the robot's functionality and reliability.

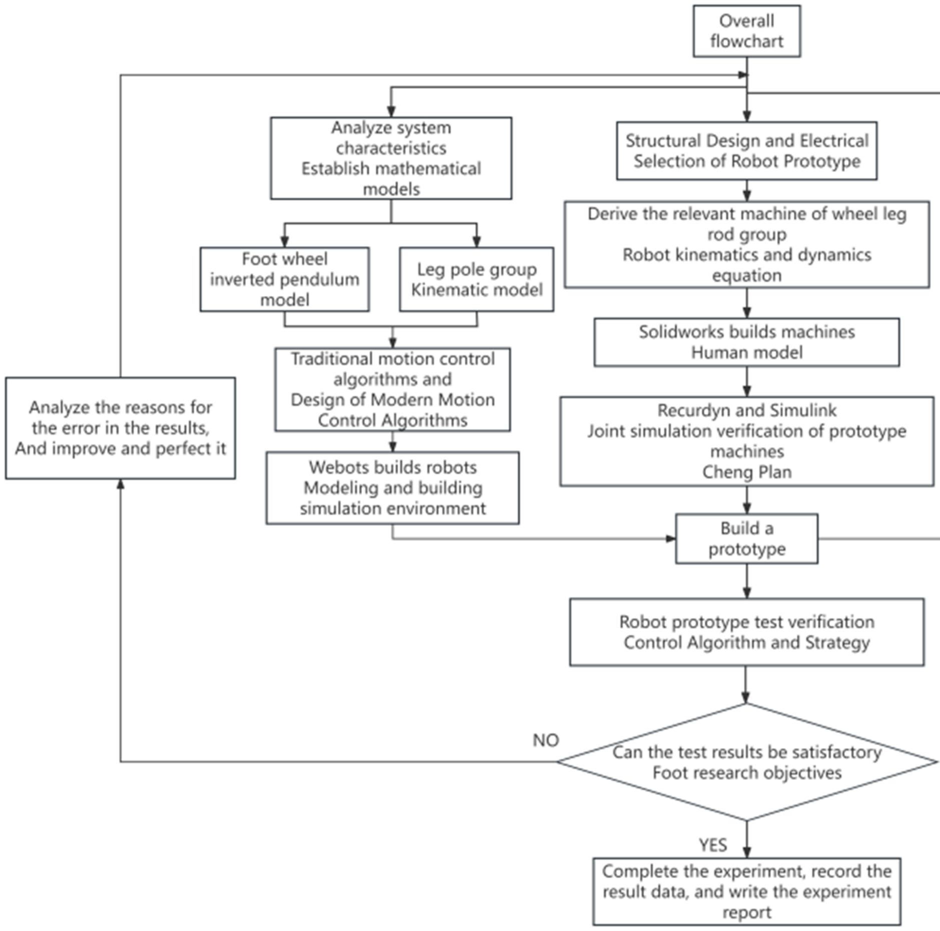

Figure 4. Program flowchart

IV. CONCLUSION

This study optimizes the power unit and develops advanced jumping trajectory planning for intelligent logistics delivery robots. It models the robot as a two-wheel balanced vehicle with electronic differential and torque coordination control, leveraging Hall sensor feedback and Field-Oriented Control (FOC). Key motor features include an axial flux structure, high peak power, and IP65 waterproof rating. Jumping trajectory planning is inspired by animal biomechanics, dividing the jump into four phases and using a virtual model for analysis. A dynamic simulation ensures alignment with engineering designs. These methods enhance the robot's versatility and adaptability in various logistics scenarios.

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